Roof Waterproofing Test Report

Recommended Processing / Fabrication Method:

1. Surface Preparation:
- Clean the concrete roof surface from dust, oil, loose particles, old weak coating, and standing moisture.
- Repair visible cracks and damaged areas before coating.
- Apply only to a dry, stable substrate unless the selected binder allows damp-surface application.

2. Powder Pre-Mix:
- Dry blend titanium dioxide or aluminum oxide reflective filler with porous silica, treated expanded perlite, or lightweight insulating filler.
- Add UV stabilizer powder and fiber reinforcement slowly to avoid clumping.
- Use hydrophobic or surface-treated insulating fillers where possible to reduce water absorption.

3. Liquid Binder Preparation:
- Prepare the acrylic or styrene-acrylic binder phase.
- Add elastomeric waterproofing additive such as SBR, acrylic elastomer, EVA, silicone, or polyurethane dispersion according to compatibility.
- Add defoamer, dispersant, and adhesion promoter if available.

4. Composite Mixing:
- Slowly add the powder blend into the liquid binder under mechanical mixing until a uniform spreadable coating is obtained.
- Avoid excessive air bubbles.
- Adjust viscosity only with compatible water or solvent recommended by the binder supplier.

5. Application:
- Apply by roller, brush, trowel, or spray depending on viscosity.
- Use at least two coats.
- Apply the first coat as a sealing/adhesion layer and the second coat as the waterproof reflective insulation layer.
- Apply the second coat after the first coat has dried sufficiently.

6. Drying and Curing:
- Allow each coat to dry fully before applying the next layer.
- Suggested ambient curing is 24?72 hours depending on temperature, humidity, coating thickness, and binder supplier guidance.
- Avoid application during rain, wet surfaces, or extreme midday heat that may cause rapid drying and cracking.

7. Quality Control:
- Check coating continuity, pinholes, adhesion to concrete, crack coverage, dry film thickness, surface defects, and water ponding resistance.
- Measure roof surface temperature reduction compared with uncoated concrete.

8. Evidence Boundary:
- This is a planning-level processing route.
- Exact mixing speed, viscosity, coat thickness, curing time, and additive levels must be optimized experimentally.
- No waterproofing, insulation, or commercial claim should be made before laboratory and rooftop validation.